



ELG 5218: Uncertainty Evaluation in Engineering Measurements and Machine Learning Final Exam (180 minutes)

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Instructions: Use only simple calculators. The total number of points is 119.

1 Theoretical Questions

(32 points)

1.1 Uncertainty Quantification

(8 points)

A model produces 200 forecast samples for a scalar target. The empirical 5th, 50th, and 95th percentiles of these samples are

$$q_{0.05} = 8, \quad q_{0.50} = 11, \quad q_{0.95} = 15.$$

The realized target value is $y = 14$.

1. Give one reasonable point prediction and state what summary you chose.
2. Give an empirical 90% prediction interval.
3. Is the realized value covered by this interval?
4. Does this single covered case prove the model is calibrated? Briefly explain.

1.2 Variational Inference and the ELBO

(8 points)

1. Why does variational inference minimize the reverse KL divergence $KL(q||p)$ rather than the forward KL divergence $KL(p||q)$ in standard formulations?
2. Consider the latent-variable model

$$p(z) = \mathcal{N}(0, 1), \quad p(x | z) = \mathcal{N}(x; z, 1),$$

with a single observed data point $x \in \mathbb{R}$. Let the variational distribution be

$$q(z) = \mathcal{N}(z; \mu, \sigma^2).$$

Write the evidence lower bound (ELBO) $\mathcal{L}(\mu, \sigma)$ as a closed-form function of μ and σ .

Here is the formula for KL divergence for two univariate Gaussians, $q(z) = \mathcal{N}(\mu, \sigma^2)$, $p(z) = \mathcal{N}(\mu_0, \sigma_0^2)$:

$$\text{KL}(q\|p) = \frac{1}{2} \left(\frac{\sigma^2}{\sigma_0^2} + \frac{(\mu - \mu_0)^2}{\sigma_0^2} - 1 + \log \frac{\sigma_0^2}{\sigma^2} \right).$$

1.3 Probabilistic Load Forecasting

(8 points)

You are building a model to forecast electrical power demand y (MW) from weather features x (temperature, humidity, hour of day). The load distribution is known to be binodal on weekends (low and high usage modes). You must provide prediction intervals alongside point forecasts.

1. Explain why a standard regression model (single point output) is insufficient for this task.
2. Would you choose a Gaussian-head (MVE) network or a Mixture Density Network? Justify your choice.
3. Describe how you would evaluate whether the model's uncertainty estimates are well calibrated, without having explicit uncertainty labels.
4. Distinguish between aleatoric and epistemic uncertainty in this forecasting problem. For an unusual weather event that is poorly represented in the training set, which type is most relevant? Why might a standard Gaussian-head model be insufficient?

1.4 Reinforcement Learning: On-Policy vs. Off-Policy

(8 points)

1. In reinforcement learning, which components of the environment model must typically be learned or inferred from interaction, whereas in planning, they are assumed to be known?
2. Define on-policy and off-policy learning.
3. Classify each of the following methods as on-policy or off-policy: SARSA, Q-learning, REINFORCE, PPO, DQN, SAC.
4. Why is off-policy learning important in practice?

2 Multi-Sensor Fusion: PDAF and Attention (25 points)

An autonomous system tracks multiple targets using radar and camera measurements. One proposed architecture uses a transformer in which track tokens attend to measurement tokens from both sensors.

1. Explain how cross-attention can be interpreted as a soft fusion or soft association mechanism between tracks and measurements.
2. Compare this with PDAF. What is the analogous quantity in PDAF, and what additional modeling assumptions does PDAF make?
3. Suppose radar and camera both produce candidate measurements for the same target. Explain how an attention-based model could combine them. What is the advantage of this learned fusion over hand-designed fusion weights?
4. A major failure mode of plain attention in multi-target tracking is the lack of a one-to-one assignment constraint. Why does this problem exist in the attention module? Describe one mechanism that addresses it.
5. Two tracks produce attention weights

$$\alpha_1 = [0.66, 0.24, 0.10], \quad \alpha_2 = [0.49, 0.46, 0.05].$$

- (a) Which track has the more ambiguous association?
- (b) Which measurement is most likely to be claimed by both tracks under hard assignment?

3 Questions Related to the Assignments

(12 points)

3.1 Bayesian Updating with a Misspecified Static Model

In Assignment 1, you examined a temperature-monitoring example in which the true temperature could drift over time rather than remain constant. Suppose the true temperature is

$$\theta_t = 22 + 0.1t,$$

but the learner still uses the standard Bayesian update rules for a model with one fixed unknown parameter θ .

1. Describe what happens qualitatively to the posterior mean and posterior variance as more observations are collected.
2. Explain why the learner may become increasingly confident while simultaneously becoming systematically wrong.
3. Does this failure mainly reflect aleatoric uncertainty, epistemic uncertainty, or model misspecification? Explain.
4. Give one Bayesian state-space or dynamic-model modification that would make tracking the drifting temperature possible.

3.2 Interpreting Predictive Bands in the Heteroscedastic Regression Model

In Assignment 2, you implemented a heteroscedastic regression model in which the observation noise was allowed to depend on the input:

$$y_i \sim \mathcal{N}(wx_i + b, \sigma(x_i)), \quad \sigma(x) = \exp(ax + c)$$

You then examined how the learned noise curve $s(x)$ changes across the input range and how this affects the predictive bands compared with the homoscedastic baseline. Recall that in this model, the predictive distribution can become wider in some regions of x either because the data are intrinsically noisier there, or because the model parameters are themselves uncertain.

A student now says:

“Because the predictive band gets wider for large x , the model is more uncertain there in every possible sense.”

Answer the following:

1. Explain why this statement confuses two different notions of uncertainty.
2. Clearly distinguish the role of the input-dependent noise function $s(x)$ from the uncertainty in the parameters (w, b, a, c) .
3. Give one concrete example of a situation in which aleatoric uncertainty is large but epistemic uncertainty is small.

4 An Integrative Question Combining Multiple Topics (30 points)

This problem examines the connections between gradient-based sampling, variational objectives, and modern diffusion-based generative models.

1. Langevin dynamics for a Gaussian target.

- (a) Derive the unadjusted Langevin algorithm (ULA) update starting from Euler-Maruyama discretization of the continuous-time Langevin SDE (formula 113) for $p(x) = \mathcal{N}(0, 1)$.
- (b) Explicitly compute the drift term and explain its effect on the dynamics.

2. Random-walk MCMC vs. Langevin dynamics.

- (a) Write the proposal distribution for Random Walk Metropolis (RWM).

- (b) Identify the key algorithmic difference between RWM and Langevin dynamics.
- (c) Explain why RWM suffers from poor scaling in high dimensions, whereas Langevin dynamics typically scales better.

3. MALA and discretization bias.

- (a) Explain how discretization bias arises in the unadjusted Langevin algorithm.
- (b) Describe how the Metropolis-Hastings correction in MALA removes this bias.
- (c) Why is a Metropolis-Hastings accept/reject step usually not used in standard diffusion-based neural samplers?

4. Diffusion models and score-based sampling. A DDPM reverse step can be written as

$$x_{t-1} = x_t + \sigma_t^2 \nabla_x \log p_t(x_t) + \sigma_t z, \quad z \sim \mathcal{N}(0, I).$$

- (a) Explain how this reverse update has the same score-driven stochastic structure as a discretized Langevin step.
- (b) State the relationship between the predicted noise ϵ and the score $\nabla_{x_t} \log p(x_t | x_0)$ in DDPMs. Why does this relationship connect DDPMs to score-based generative modeling?

5. Why diffusion uses multi-scale noise instead of plain Langevin.

- (a) Explain why single-scale Langevin dynamics fails on complex or multimodal data distributions.
- (b) Explain how the forward noising process in diffusion models alleviates this issue.

6. Connection to ELBOs and variational learning.

- (a) Briefly explain how score matching in diffusion models relates to variational objectives such as the ELBO.

5 Interpretation of Model Results

(20 points)

In this question, a *safety-critical application* refers to a setting where incorrect or overconfident predictions can lead to harm, failure, or unsafe decisions. Examples include monitoring infrastructure, medical forecasting, autonomous systems, or early-warning systems.

A paper reports the following four models on a forecasting task:

Model	RMSE	NLL	90% Coverage	CRPS
A	0.50	2.40	0.58	0.41
B	0.58	1.10	0.89	0.25
C	0.47	3.10	0.42	0.55
D	0.63	1.35	0.96	0.31

1. Which model is most appropriate for a safety-critical application? Justify your answer using at least two metrics, explaining why those metrics matter in this context.
2. If only RMSE were reported, which model would appear best? Explain why relying solely on RMSE can be misleading in safety-critical or uncertainty-aware applications.
3. Which model appears best calibrated with respect to the nominal 90% prediction interval? Briefly explain what undercoverage and overcoverage indicate.
4. Based on the table, which model demonstrates the best overall probabilistic forecasting performance? Your explanation should reference the role of both sharpness and calibration.

When answering the questions, explicitly refer to the definitions of the metrics and the intuition behind what each metric measures (accuracy, calibration, uncertainty, or distributional quality), not just their numerical values.